

IP-SAKTI SAHAYAK

Family 4 — Zingiberaceae / Haldi (Turmeric)

Objective: Build a TKDL-only family cluster around *Curcuma longa* so IP-SAKTI can identify closely related TKDL-listed plants and rank them as potential alternatives for further research when relevant prior-art/IP risk is detected.

Strict source rule: All five plant candidates below appear in the official TKDL plant-name index. No outside plant source has been used to add candidates. The public TKDL material reviewed provides strong formulation/IP evidence for *Curcuma longa* and *Curcuma zedoaria*; for the other candidates, the index confirms TKDL presence but a specific matching formulation is not claimed here unless a public TKDL record was located.

1. Anchor / main plant

Haldi / Turmeric — *Curcuma longa*

TKDL Ayurvedic terminology includes Haridra. TKDL also records *Curcuma longa* in Unani material.

2. Related TKDL species — 5 candidates

TKDL plant	Relation to Haldi	Prototype similarity	What model should do
<i>Curcuma amada</i> Roxb.	Same genus; Zingiberaceae	HIGH — 0.88	Prioritize; compare part/use/formulation evidence
<i>Curcuma angustifolia</i> Roxb.	Same genus; Zingiberaceae	HIGH — 0.84	Prioritize; compare part/use/formulation evidence
<i>Curcuma aromatica</i> Salisb.	Same genus; Zingiberaceae	HIGH — 0.88	Prioritize; compare part/use/formulation evidence
<i>Curcuma caesia</i> Roxb.	Same genus; Zingiberaceae	HIGH — 0.84	Prioritize; compare part/use/formulation evidence
<i>Curcuma zedoaria</i> (Christm.) Roscoe	Same genus; Zingiberaceae	VERY HIGH — 0.92	Strongest initial alternative candidate; direct TKDL formulation/IP evidence available

Similarity scores are prototype ranking values, not scientific equivalence or legal clearance.

3. TKDL evidence for the anchor and candidate pool

Official TKDL plant-name index: <https://www.tkdil.res.in/tkdil/langdefault/common/Utility/Keyword/Plant-Name.asp>

The official TKDL index lists *Curcuma amada*, *Curcuma angustifolia*, *Curcuma aromatica*, *Curcuma caesia*, *Curcuma longa*, *Curcuma pseudomontana* and *Curcuma zedoaria*. This gives us a larger TKDL-only candidate pool; this PDF uses five of those entries for the first model cluster.

1782/DEL/2007: TKDL prior-art material describes *Curcuma longa* (Haridra/turmeric) and *Zingiber officinale* in a therapeutic composition, with turmeric rhizome powder listed as an ingredient. The report states the herbal composition was known in the prior art and concluded the claims lacked novelty and inventive step; the application was abandoned.

Official TKDL record: https://www.tkdil.res.in/tkdil/LangGerman/common/Examinarreport/reports/1782_DEL_2007.asp

1066/DEL/2009: TKDL pre-grant opposition cited *Curcuma longa* for cough/bronchitis, bronchial asthma and throat inflammation. The examination report cites several TKDL formulations, including Kumaryasava and Draksa Avaleha, containing *Curcuma longa* in combinations with other ingredients; the examiner considered the claimed combinations in view of prior art.

Official TKDL record: https://www.tkdil.res.in/tkdil/LangFrench/common/Examinarreport/reports/1066_DEL_2009.asp

EP1883416: TKDL third-party observations brought prior-art references on *Curcuma longa* for bronchial asthma, inflammation, psoriasis and rheumatism. The cited prior-art material included Ayurveda and Unani texts; the examination report states the application was abandoned after TKDL evidence and other cited documents.

Official TKDL record: <https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/EP1883416.asp>

372/KOLNP/2009: TKDL pre-grant opposition lists *Curcuma longa* and *Curcuma zedoaria* among plants used in traditional medicine for immune-suppressive/emaciating disease, diseases of the abdomen, stomatitis, mouth diseases, gastritis and related uses. The record identifies multiple Ayurveda/Siddha/Unani sources.

Official TKDL record: https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/372_KOLNP_2009.asp

1865/DEL/2006: The official TKDL examination PDF includes *Curcuma zedoaria* (Tumiri/Kachuri/Zarambad) as an ingredient in a multi-ingredient therapeutic composition. This provides direct formulation-level evidence for the candidate species.

Official TKDL PDF: https://tkdil.res.in/tkdil/LangFrench/Common/Examinarreport/pdf_files/1865-DEL-2006.pdf

201721000834 — TKDL outcome/claims material: The official TKDL outcome material includes a herbal preparation containing *Curcuma amada* in a claimed formulation for an anti-osteoporotic preparation. This supports treating *Curcuma amada* as more than a bare plant-index entry in the IP-SAKTI corpus.

Official TKDL material: https://www.tkdil.res.in/tkdil/Outcomes/201721000834_Amendedclaims.pdf

4. Similarity engine — how to suggest an alternative

Step 1: Detect the user's anchor plant (e.g., *Curcuma longa*). **Step 2:** resolve its TKDL botanical identity and family. **Step 3:** retrieve only TKDL-listed candidates from that family/genus. **Step 4:** compare plant part, traditional indication, formulation role and specific TKDL evidence. **Step 5:** rank and explain the candidate.

Recommended ranking: Same genus + documented formulation/use overlap = highest priority. Same genus with only plant-index evidence = provisional candidate. Same family but different genus = lower priority. Any candidate with conflicting identity/evidence = flag for human review.

Example UI: “*Curcuma zedoaria* — Very High similarity candidate. Same genus as turmeric and supported by a public TKDL formulation record. Investigate further; this does not establish that the substitute avoids patent infringement.”

5. Model-ready dataset fields

family | anchor_plant | candidate_plant | botanical_name | TKDL_name/synonyms | plant_part | traditional_use | formulation | formulation_role | source_text | TKDL_case_id | official_TKDL_url | similarity_score | similarity_reasons | patent_IP_flag | evidence_level | verification_status

6. AI / legal guardrails

- Same family/genus ≠ automatically interchangeable.
- Same family/genus ≠ freedom from patent infringement.
- Never output “no legal barriers” without a separate claim-level patent analysis.
- Distinguish TKDL traditional-knowledge evidence from novelty, inventive step and Section 3(p).
- Show the exact official TKDL record behind every high-confidence recommendation.
- Final wording: “**Potential alternative candidate for further research — botanical, formulation, efficacy and patent/legal verification required.**”

7. Verification note

This family intentionally uses five candidates that are present in the official TKDL plant index. The similarity score is an IP-SAKTI prototype feature and should be recalibrated after the team extracts the underlying TKDL formulation records. The public index itself should not be treated as proof of formulation equivalence.